Naïve Bayes Classification Summary

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Source

CMP 466 - Machine Learning and Data Mining

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Structure Overview

1. Bayes Theorem

- Basic probability framework for classification
- P(A|B) = P(B|A) * P(A) / P(B)

2. Using Bayes for Classification

- Classifying instances into categories based on features
- Naïve Bayes applies this theorem to classification problems

3. Naïve Bayes Classifiers

- Core Assumption: Features are conditionally independent given the class
- Why "Naïve"? Assumes words/features don't influence each other (often unrealistic but computationally efficient)

4. Implementation in SKLearn

- Practical implementation using scikit-learn library

Text Classification Examples

Example 1: Single Word Classification

- Scenario: Classifying emails as from Adam or Lena
- Word: "Love"
- Calculation:
- P(Adam | "Love") = 0.04 / (0.04 + 0.03) = 0.57
- P(Lena | "Love") = 0.03 / (0.04 + 0.03) = 0.43
- Result: Adam (57% probability)

Example 2: Multiple Features

- Scenario: Adding "Love Life" features
- Shows how multiple independent features are combined

Example 3: Posterior Probability with Normalization

- Demonstrates proper probability calculation
- Ensures probabilities sum to 1 across classes

Example 4: Word Order Independence

- Key Insight: Word order doesn't matter in Naïve Bayes
- Only the presence/absence of words counts
- Commutative property: P(Adam | "Life Deal") = P(Adam | "Deal Life")

Core Calculation Method

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P(Class | Features) = P(Features | Class) * P(Class) / P(Features)
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Since features are assumed independent:

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P(Features | Class) = P(feature1 | Class) P(feature2 | Class) ...
```

Key Takeaways

1. Naïve Bayes is efficient for text classification due to independence assumption
2. Works well even when assumption is violated in practice
3. Computationally simple compared to other classifiers
4. Probabilistic framework provides uncertainty estimates

Limitations

- Independence assumption rarely holds in real data
- May underperform on complex feature interactions
- Sensitive to class priors and feature distributions

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